

Informational Power Through Pivotality: When a Hegemon May Choose Consensus

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Abstract

When does a powerful state prefer consensus to majority rule in an international organization? I develop a formal model of institutional design in which a hegemon has private information about the value of multilateral cooperation. Under majority rule, weaker states can form winning coalitions without the hegemon, so its private information has limited strategic value. Under unanimity, every agreement requires the hegemon's approval, forcing weaker states to decide how much to offer without knowing how valuable cooperation really is. This uncertainty works in the hegemon's favor: weaker states pay more than necessary to secure agreement, and majority rule eliminates this advantage entirely. In a calibration to OPEC circa 1986, the hegemon captures 28% of the cooperation surplus under unanimity (9 percentage points above its outside option) compared to 23% under majority (4 points above), a screening premium of 5 percentage points that majority rule eliminates entirely. This screening advantage means that, once cooperation is promising enough for weaker states to participate under unanimity, the hegemon strictly prefers it. Consensus, in this account, is not a concession to weaker states but an institutional arrangement that makes informational advantage politically productive—offering a distributive explanation for consensus rules in organizations such as OPEC.

1 Introduction

Most international organizations operate by consensus, even when powerful states drive their creation (Gould 2016). The international arena is where material asymmetries should matter most relative to domestic politics. If institutional rules reflect power, powerful states might be expected to favor arrangements that let them turn that power into a distributive advantage. Why would a powerful state ever prefer consensus over majority rule, where it could use agenda power to impose its preference without any state having veto power?

Under majority rule, a powerful actor can form coalitions, exclude rivals, and keep the surplus (Baron and Ferejohn 1989; Kalandrakis 2006; Eraslan and Evdokimov 2019). Consensus does the opposite: by giving every participant a veto, it forces powerful states to surrender the tools they might value most. Yet international organizations are built around consensus, even when powerful states drive their creation (Gould 2016; Steinberg 2002; Koremenos, Lipson, and Snidal 2001). Existing accounts usually see consensus either as constraining powerful states through institutional self-binding (Keohane 1984; Ikenberry 2001) or as a façade for informal power that operates irrespective of the formal rule (Steinberg 2002; Stone 2011; Gruber 2000). Neither explains when and why a hegemon would prefer unanimity to a rule that allows exclusion.

This paper theorizes when and why a hegemon may prefer unanimity to majority rule in international organizations. The central claim is that consensus is an institutional technology of power. I show that consensus can make informational power more politically productive than the coalition arithmetic of majority rule. By informational power I mean the bargaining advantage that a privately informed actor derives from being pivotal: when other players must secure its approval without knowing its type, uncertainty itself becomes a source of rent.

To formalize the argument, I develop a model in which a hegemon chooses between majority and unanimity. Proposal rights are symmetric under both rules, isolating the effect of the voting rule itself. The hegemon privately observes the value of cooperation before weaker states decide whether to enter. The central difference is that the two rules reward this informational advantage differently. Under majority rule, weak states can form coalitions without making the hegemon pivotal, eliminating the screening problem. Under unanimity, weak states must include the hegemon in every proposal and decide how much to offer without knowing the probability of acceptance. This

screening generates a cutoff at which weak proposers switch from aggressive to conservative treatment of the hegemon, giving the hegemon a strictly higher conditional payoff than under majority at every level of uncertainty. The institutional comparison reduces to a single distinction: unanimity dominates wherever it can sustain entry, and majority's only advantage is wider institutional viability.

The substantive implication is that consensus is most valuable to a hegemon when the prospects of multilateral cooperation are promising enough that weaker states are willing to come to the table. Under those conditions, unanimity amplifies the hegemon's informational advantage. On the other hand, when cooperation looks too uncertain for weaker states to participate at all, majority rule can be preferred because it lowers the bar for participation.

2 Consensus, Information, and Institutional Design

The dominant explanations for consensus in international organizations fall into two categories, each missing a different piece of the puzzle. Self-binding accounts hold that powerful states accept formal constraints to make credible commitments, encouraging weaker partners to participate (Keohane 1984; Ikenberry 2001). These accounts explain why consensus sustains cooperation, but not why it confers a distributive advantage on the hegemon. Informal power accounts imply that agenda power, control over drafting, and asymmetric resources enable powerful states to shape outcomes regardless of formal rules (Steinberg 2002; Stone 2011; Gruber 2000). They cannot explain why formal rules vary systematically across organizations. Gould (2016) documents the prevalence and variation of consensus comprehensively. This paper identifies a mechanism absent from existing accounts: consensus activates informational power. The hegemon's private information is more valuable under unanimity than under majority rule.

The model builds on Baron-Ferejohn's legislative bargaining, where proposal rights drive political power (Baron and Ferejohn 1989; Kalandrakis 2006; Eraslan and Evdokimov 2019). Holding proposal rights symmetric isolates the effect of the voting rule. The departure is combining asymmetric information about the value of cooperation with a choice between voting rules. Under unanimity, the combination generates endogenous screening; under majority, it does not.

The most closely related papers to the present one are Koremenos, Lipson, and Snidal (2001) and Kim, Kim, and Van Weelden (2025). The paper by Kim, Kim, and Van Weelden (2025) examines persuasion in veto bargaining, analyzing how an information designer augments a proposer’s agenda-setting power when a veto player is present. Their model is bilateral and takes the veto rule as given. The article by Koremenos, Lipson, and Snidal (2001) conjectures (though it does not formalize them) that powerful states seek asymmetric control, while uncertainty increases demand for unanimity as insurance for risk-averse states. The present model shows that uncertainty indeed makes the hegemon’s private information valuable, and there is no need to appeal to risk aversion¹.

The informational mechanism generates a discriminating prediction. Because screening rents require genuine uncertainty about the value of cooperation, the model predicts a negative cross-sectional association between consensus rules and transparent distributive stakes. Financial institutions, where capital contributions and loan terms make the pie calculable, should not adopt the consensus rule and instead use weighted majority. Regulatory and standard-setting bodies, where consequences are difficult to forecast, should adopt consensus most frequently (Gould 2016). Rational design accounts also predict unanimity under uncertainty (Koremenos, Lipson, and Snidal 2001), but through risk aversion rather than informational power. The two channels differ in their distributional implications: insurance protects all members symmetrically, while screening benefits the privately informed. Legitimacy accounts do not condition on informational structure. Self-binding accounts predict consensus where commitment problems are severe, regardless of transparency. Informal power accounts cannot explain why consensus is absent precisely where powerful states have the strongest informal leverage.

3 A Motivating Example

Before turning to the general model, consider a simplified special case with three players and one bargaining round to build intuition for the mechanism. The general model (Section 4) extends this to N players, two rounds of Baron-Ferejohn bargaining, and entry costs. There is one hegemon H and two weak states W_1, W_2 . The value of cooperation is either $V(0) = 1$ (low) or $V(1) = 2$ (high); H observes the state, the weak states do not. Each weak state’s outside option is 0; H ’s

¹For information design under unanimity without bargaining, see Bardhi and Guo (2018).

outside option is $\alpha V(\theta)$ with $\alpha = 0.2$. A proposer is selected uniformly at random; to illustrate the screening mechanism, I focus on the case where a weak state is selected and makes a take-it-or-leave-it offer specifying H 's share y_H . If H rejects, each player receives their disagreement payoff.

Throughout the paper, I model consensus as unanimity over a binary accept/reject decision. This abstracts the distinction between unanimity (everyone must explicitly consent to an agreement) and consensus (no one explicitly blocks a deal), but captures the central strategic feature: any member can block agreement.

Under majority rule, W_1 and W_2 can pass a proposal without H 's vote. The proposing weak state excludes H from the coalition, and H captures $0.2V(\theta)$ from its bilateral alternative regardless of what W believes. The hegemon's expected payoff is $0.2 + 0.2\mu$, where $\mu = \Pr(\theta = 1)$. The payoff is linear in beliefs, with no strategic discontinuity. Because the hegemon's vote is never needed, its private information creates no leverage.

Under unanimity, H 's vote is required, and the proposing weak state faces a dilemma. It must decide how much to offer H without knowing θ :

- An aggressive offer ($y_H = 0.2$) pays the hegemon its low-state outside option. Type $\theta = 0$ is indifferent and accepts; type $\theta = 1$ rejects, since it wants 0.4. The weak proposer gets 0.8 if $\theta = 0$ and 0 if $\theta = 1$.
- A conservative offer ($y_H = 0.4$) pays the hegemon its high-state outside option. Both types accept. The weak proposer gets 0.6 if $\theta = 0$ and 1.6 if $\theta = 1$.

The weak state prefers the aggressive offer when $0.8(1 - \mu) > 0.6(1 - \mu) + 1.6\mu$, which simplifies to $\mu < 1/9$. This defines a screening cutoff: the belief at which the weak proposer switches from aggressive to conservative treatment of the hegemon.

The switch in behavior creates a discrete jump in the hegemon's expected payoff. Below the cutoff, the hegemon earns $0.2 + 0.2\mu$, the same as under majority. Above it, the expected payoff jumps to 0.4, constant and independent of μ . The source of the jump is overpayment: on the conservative branch, type $\theta = 0$ receives 0.4 but would accept 0.2. Weak states cannot avoid this cost because they must secure the hegemon's vote without knowing the type. At the cutoff $\mu^* = 1/9$, the jump

is $8/45 \approx 0.18$, about 16% of expected surplus.

The screening jump gives the hegemon a strictly higher conditional payoff under unanimity than under majority for every belief above the cutoff. The general model (Section 4 onwards) shows this holds across the entire belief space (Theorem~1).

The general model enriches this logic in several respects. With N players and two-round Baron-Ferejohn bargaining with discounting ($\beta < 1$), the richer environment produces two screening cutoffs (one per round) and an entry threshold that creates a second channel through which the voting rule matters. Theorem~1 shows that the pattern identified here generalizes: conditional on entry, unanimity gives the hegemon a strictly higher payoff at every level of uncertainty. The institutional classification (Proposition~4) establishes that unanimity dominates wherever it can sustain entry; majority dominates only through wider institutional viability.

4 The Model

Definition 1 (Institutional Design Game). *The game $\Gamma(N, p, r, \alpha, \beta, c)$ has the following primitives.*

- *There are $N \geq 3$ players: one hegemon H and $N - 1$ weak states $W_1, \dots, W_{N-1}$.*
- *Nature draws $\theta \in \{0, 1\}$ with prior $\Pr(\theta = 1) = p \in (0, 1)$.*
- *The state determines the value of multilateral cooperation: $V(0) = 1$ and $V(1) = r > 1$.*
- *The hegemon observes θ ; the weak states do not.*
- *Disagreement payoffs: each W receives 0;² H receives $\alpha V(\theta)$, with $\alpha \in (0, 1/r)$.*
- *All players are risk-neutral: payoffs are linear in shares of $V(\theta)$.*
- *The common discount factor is $\beta \in (0, 1)$.*
- *Each weak state that joins the institution pays an entry cost $c > 0$.*
- *The voting rule is $R \in \{M, U\}$: under majority (M), decisions pass with $q = \lfloor N/2 \rfloor +$*

²The normalization $d_W = 0$ is without loss of generality: it requires only that weak states' outside options are symmetric and strictly lower than the hegemon's. The pie $V(\theta)$ is then interpreted as the surplus above weak states' common outside option.

1 votes; under unanimity (U), decisions require all N votes.³ Both rules have symmetric proposal rights (each player recognized with probability $1/N$).

Let $V_e(\mu) \equiv 1 + \mu(r - 1)$ denote the expected value of cooperation at posterior belief μ , and define $x \equiv (N - 1)\alpha r$ as a notational shorthand.

The game proceeds in three stages. In Stage 0, the hegemon chooses the voting rule $R \in \{M, U\}$. In Stage 1, nature draws θ (observed by H only); weak states simultaneously decide whether to enter, paying cost c . The institution forms if and only if all $N - 1$ weak states enter.⁴ In Stage 2, members play a two-round Baron-Ferejohn bargaining game:⁵ in each round, a proposer is selected uniformly at random, offers a division of $V(\theta)$, and all members vote under rule R . Rejection in Round 1 leads to Round 2 with payoffs discounted by β ; rejection in Round 2 yields disagreement payoffs.

The solution concept is Perfect Bayesian Equilibrium (PBE).⁶ Each weak state's information sets pool the two branches of θ : at any decision node, W_i knows the history of proposals and responses, but not θ . Weak states hold belief μ over θ , which equals the prior p at the entry stage and may be updated during bargaining (e.g., after a rejection in Round 1). This asymmetry is what generates the screening problem when a weak state proposes under unanimity.

Figure 1 presents the extensive form of the bargaining subgame under unanimity (Stage 2).

³In this game, each player's action is binary: accept or reject. There is no abstention or silence option. The institutional distinction between consensus—where silence is treated as tacit approval—and formal unanimity—where explicit approval is required—therefore does not arise: with binary actions {accept, reject}, the two rules are equivalent. I use the terms interchangeably. In practice, “consensus” at the WTO refers to the norm of not calling a formal vote, which approximates unanimity when any member can block by objecting.

⁴The all-or-nothing entry assumption and the interpretation of N are discussed in the Scope section.

⁵Two rounds ensure that rejection leads to a continuation game rather than immediate disagreement, so the screening mechanism does not depend on ultimatum structure. The motivating example (Section [refexample](#)) shows it operates in one round; two rounds confirm robustness in the Baron-Ferejohn environment.

⁶Tie-breaking convention: when H is indifferent between accepting and rejecting an offer (offer equals reservation value), H accepts.

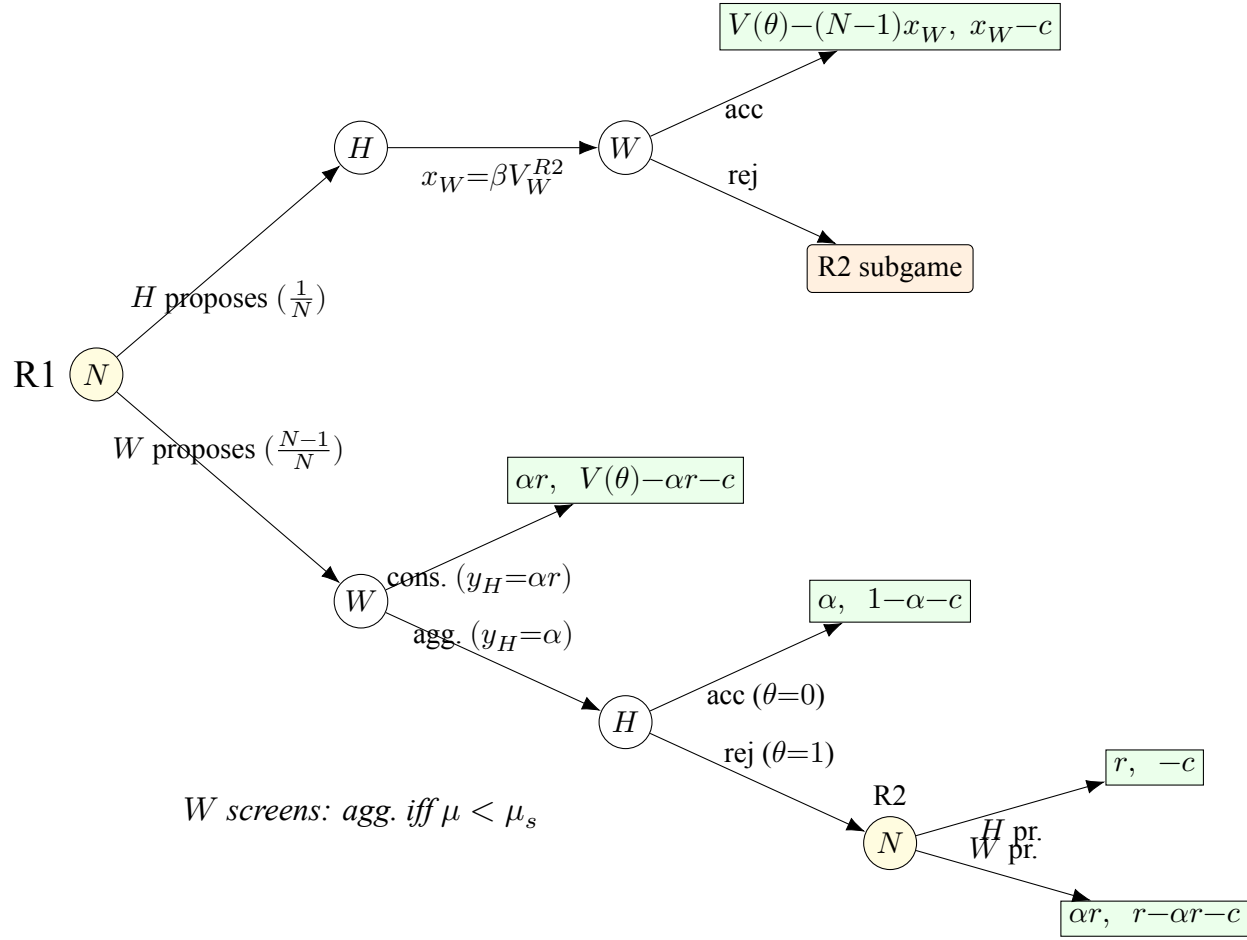


Figure 1: Extensive form: Stage 2 (bargaining under unanimity, Rounds 1-2). Payoffs are (u_H, u_W) . A proposer is selected uniformly (1/N). When W proposes, W chooses between an aggressive offer that only $\theta=0$ accepts and a conservative offer that both types accept. Under the aggressive offer, $\theta=1$ rejects, revealing the state and leading to R2 (payoffs discounted by β). In R2, W knows $\theta=1$ (complete information); W offers αr , H offers 0; disagreement yields $(\alpha r, -c)$. When H proposes in R1, H offers βV_W^{R2} to each W ; W accepts in equilibrium. Offer labels and terminal payoffs show R2 values (where H 's reservation is $\alpha V(\theta)$) for readability; exact R1 offer levels are $\beta V_H^{R2}(\theta=0, \mu'=1)$ (aggressive) and $\beta V_H^{R2}(\theta=1)$ (conservative), derived in Appendix A.3. Under majority rule, H 's vote is not pivotal, eliminating the screening mechanism (Proposition 1).

5 Majority Rule: Linear Bargaining Without Screening

Under majority rule, a proposal passes with $q = \lfloor N/2 \rfloor + 1$ votes. The key structural feature is that weak states can form a winning coalition without including the hegemon. When a weak state proposes, it assembles a majority from the other weak states alone; the hegemon, excluded from the coalition, captures its bilateral alternative $\alpha V(\theta)$ regardless of what weak states believe about the state of the world.

Proposition 1 (Majority rule produces no screening). *Under majority rule with symmetric proposal rights, the hegemon's expected continuation value $E_\theta[V_H^{R1}(\theta, \mu, M)]$ is affine in posterior beliefs μ . There is no screening cutoff.*

Proof. See Appendix B.1. □

Because the hegemon's vote is never needed, weak states never face a screening problem, and the hegemon earns no screening rent. Majority transforms bargaining into coalition arithmetic: the proposer buys the cheapest votes, and the hegemon is simply paid its outside option.

6 Consensus: Screening Through Pivotal Inclusion

Under unanimity, every proposal requires the approval of all members, including the hegemon. When a weak state proposes, it must secure H 's vote without knowing whether the pie is high or low. As in the motivating example (Section 3), this creates a screening problem: the proposer must choose between an aggressive offer that only the low type accepts and a conservative offer that both types accept. The conservative offer overpays the low type, which is the cost of uncertainty.

This screening logic operates in both bargaining rounds. In each round, the weak proposer's indifference between aggressive and conservative offers defines a screening cutoff: a belief threshold above which the proposer switches from aggressive to conservative treatment. The R2 cutoff has a closed form in all cases; the R1 cutoff has a closed form when the hegemon's outside option is moderate (Appendix A).

Proposition 2 (R1 screening cutoff under consensus). *Under unanimity, there exists a unique screening cutoff $\mu_s^{R1} \in (0, 1)$ such that the weak proposer prefers the aggressive offer for $\mu < \mu_s^{R1}$*

and the conservative offer for $\mu > \mu_s^{R1}$. When $\alpha < \bar{\alpha}(r, \beta, N)$, the cutoff takes the closed form:

$$\mu_s^{R1} = \frac{\phi - \sqrt{\phi^2 - 4(N-2)}}{2(N-2)}, \quad \phi \equiv \frac{rN - \beta(N-1+r)}{\beta(r-1)}, \quad (1)$$

and is independent of α . The regime threshold is

$$\bar{\alpha}(r, \beta, N) \equiv \frac{\mu_s^{R1} \cdot r}{r-1 + \mu_s^{R1}}, \quad (2)$$

where μ_s^{R1} is evaluated from (1). For $\beta = 1$, the cutoff simplifies to $\mu_s^{R1} = 1/(N-2)$.⁷

Proof. See Appendix B.2. □

The independence of the screening cutoff from the hegemon's outside option strength, which holds when $\alpha < \bar{\alpha}$ —is substantively important. What determines how aggressively weak states treat the hegemon is not how valuable bilateral alternatives are, but how much the state of the world affects the gains from cooperation, that is, the dispersion $r - 1$. The cutoff also depends on bargaining patience and the number of states that must agree.

Proposition 3 (Screening creates an informational rent). *Under unanimity, the hegemon's expected continuation payoff in R1 has an upward jump at μ_s^{R1} :*

$$Jump = (1 - \mu_s^{R1}) \cdot \frac{(N-1)\beta(r-1)}{N^2} > 0. \quad (3)$$

Proof. See Appendix B.3. □

At the screening cutoff, weak proposers switch from treating the hegemon as the low type to hedging against the possibility of the high type. The conservative offer pays the low type as if it could be the high type, creating a discrete upward shift in the hegemon's expected payoff. The screening rent is largest when the dispersion in cooperation value is high (large r), when bargaining is patient enough to make the continuation threat credible (large β), and in smaller organizations, where the

⁷When $\alpha \geq \bar{\alpha}$, the R1 cutoff falls on the low R2 branch and depends on α (Appendix A.5). The cutoff remains unique and the screening jump remains positive. The proof of Theorem 1 (Appendix B.5) covers both regimes explicitly: the payoff decomposition is additive in the R1 and R2 corrections regardless of the relative ordering of μ_s^{R1} and μ_s^{R2} , and the endpoint argument establishes $D(\mu) > 0$ on all branches. The Corollary and Proposition then follow from Theorem 1 without further dependence on the regime.

per-member screening advantage $(N-1)/N^2$ is larger (the expression is monotonically decreasing in N for $N \geq 2$).

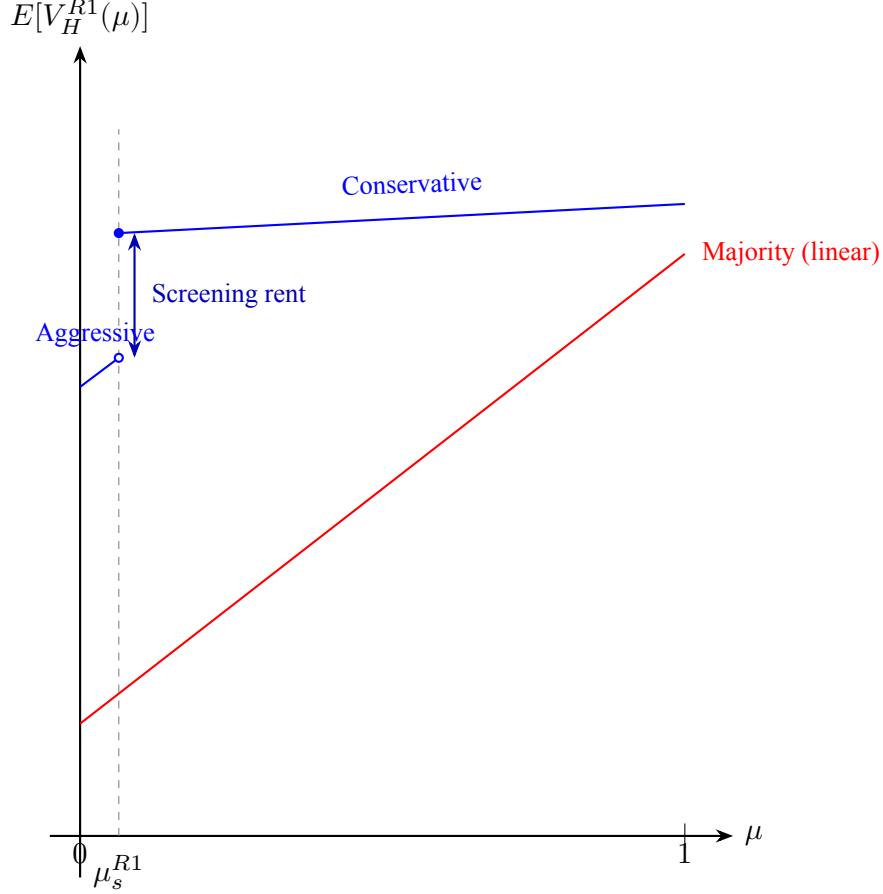


Figure 2: Schematic of the hegemon's expected R1 payoff as a function of posterior belief μ , based on Example 1 parameters ($N = 13$, $r = 1.5$, $\alpha = 0.19$, $\beta = 0.9$). Under majority (red), the payoff is linear in μ with no discontinuity. Under unanimity (blue), the payoff has two branches separated by an upward jump at the screening cutoff $\mu_s^{R1} \approx 0.064$: to the left, weak proposers play aggressively; to the right, they switch to conservative offers that overpay the low type. The gap is the screening rent that drives the hegemon's preference for unanimity (Theorem 1).

Example 1 (Screening jump: concrete magnitudes). Let $N = 13$, $r = 1.5$, $\alpha = 0.19$, $\beta = 0.9$.⁸

⁸The parameters are calibrated to OPEC circa 1986 (Section 9.1). $N = 13$ is the number of OPEC members. $r = 1.5$ reflects the ratio of cartel-sustained to competitive oil prices (\$12 to \$8 per barrel in the 1986–87 period). $\alpha = 0.19$ is Saudi Arabia's outside-option share, computed as the difference between Saudi Arabia's market share in the non-cooperative equilibrium ($\approx 25\%$) and the average share of a remaining member ($\approx 6\%$), following the normalization $d_W = 0$ [griffin1994oil]. $\beta = 0.9$ reflects moderate discounting, standard in Baron-Ferejohn

The R1 screening cutoff is $\mu_s^{R1} \approx 0.064$. At beliefs just below the cutoff, weak proposers play aggressively and the hegemon's expected payoff is $E[V_H^{R1}] \approx 0.315$. Just above the cutoff, the payoff jumps to ≈ 0.345 —an increase of about 2.9% of expected surplus. Under majority at the same belief, $E[V_H^{R1}] \approx 0.234$, with no jump anywhere. The screening mechanism gives the hegemon 35% more than majority on the aggressive branch and 48% more on the conservative branch. The discrete gap between branches is the screening rent that drives the hegemon's preference for unanimity.

7 Entry and Institutional Viability

Having established that unanimity generates a conditional payoff advantage for the hegemon (Propositions 2–3), I now turn to the entry margin, the channel through which majority can nonetheless dominate.

A weak state enters the institution if its expected bargaining payoff exceeds the entry cost $c > 0$.⁹ Because unanimity gives the hegemon a higher payoff conditional on participation (as shown in Section 8 below), it leaves less surplus for weak states, making entry harder: $\tau(U) \geq \tau(M)$.

When the institution does not form, the hegemon receives $\alpha V_e(\mu)$ from its bilateral alternative. Since this outside-option payoff is affine in μ and common to both rules, the institutional comparison depends only on the gain from bargaining relative to this baseline.

Definition 2 (Net gain function). *The hegemon's net gain from the institution under rule R at posterior μ is*

$$v(\mu, R) \equiv \begin{cases} E_\theta[V_H^{R1}(\theta, \mu, R)] - \alpha V_e(\mu) & \text{if weak states enter,} \\ 0 & \text{otherwise.} \end{cases}$$

Under majority, $v(\mu, M)$ is affine above the entry threshold. Under unanimity, the screening jump

applications.

⁹The entry cost c may represent fixed costs of negotiating accession, per-round costs of preparing delegations, or political costs of committing to multilateral disciplines. In established organizations, c is best interpreted as the cost of *substantive engagement*: allocating scarce diplomatic and analytical resources to a particular negotiating round. Since $V(\theta)$ is normalized independently of N , per-capita payoffs scale as $O(1/N)$. For large- N applications, c should be read as a per-capita cost: writing $c = \tilde{c}/N$, the entry condition becomes $N \cdot V_W \geq \tilde{c}$, where $N \cdot V_W$ is $O(1)$. All equilibrium cutoffs (μ_s^{R1} , μ_s^{R2} , α^*) are homogeneous of degree zero in the surplus and thus independent of this normalization.

at μ_s^{R1} creates a discontinuity that gives the hegemon a higher net gain at every belief where entry occurs (Theorem~1).

8 Institutional Choice

This section establishes the paper's main result: conditions under which the hegemon prefers unanimity.

8.1 Conditional payoff comparison

Before turning to the full institutional comparison, I establish the paper's central result: whenever the hegemon's bilateral alternatives are not too valuable, unanimity gives the hegemon a strictly higher expected bargaining payoff than majority at every prior belief.

Define the *conditional dominance threshold*

$$\alpha^*(N, \beta) \equiv \frac{\beta(q-1)}{N(N-1) - \beta(N^2 - N - q + 1)}, \quad q = \lfloor N/2 \rfloor + 1. \quad (4)$$

The parameter α measures the strength of the hegemon's bilateral outside option relative to multilateral cooperation. The threshold α^* marks the point at which the cost of buying unanimous agreement ($N - 1$ votes, versus $q - 1$ under majority) outweighs the screening advantage. Below α^* , the screening rent from unanimity exceeds the higher vote-buying cost at every prior belief. Above α^* , the screening rent no longer covers the cost near $\mu = 1$, where uncertainty is low and vote-buying costs dominate.

Theorem 1 (Conditional payoff dominance of unanimity). *The condition $\alpha < \alpha^*(N, \beta)$ holds if and only if, for every $\mu \in (0, 1]$,*

$$E_\theta[V_H^{R1}(\theta, \mu, U)] > E_\theta[V_H^{R1}(\theta, \mu, M)].$$

Proof. See Appendix B.5. □

Because unanimity's vote-buying cost grows with N while majority's grows more slowly, α^* de-

creases with organization size: for $\beta = 0.9$, $\alpha^* \approx 0.47$ when $N = 5$ but falls to ≈ 0.03 when $N = 164$. The condition is easy to satisfy in small institutions but demanding in large ones. For a WTO-sized organization, conditional dominance requires that the hegemon's bilateral alternatives capture only a small fraction of the multilateral surplus. The screening jump persists regardless of α , but in large organizations unanimity may not dominate at every belief. Section 9.2 discusses what happens when $\alpha \geq \alpha^*$.

Remark 1 (Robustness beyond α^*). *When $\alpha \geq \alpha^*$, the conditional advantage of unanimity fails only near $\mu = 1$. On the conservative branch ($\mu > \mu_s^{R1}$), the payoff difference decomposes as*

$$D(\mu) = \underbrace{D(1)}_{\text{without info}} + (1 - \mu) \cdot \underbrace{\Gamma}_{\text{value of uncertainty}},$$

where $C_{buy} \equiv \beta(q - 1)(1 - \alpha)$ is the vote-buying cost under majority, $C_{out} \equiv N(N - 1)\alpha$ is the aggregate outside-option cost, $D(1) = r[C_{buy} - C_{out}(1 - \beta)]/N^2 \leq 0$ is the payoff comparison when the hegemon's type is known, and $\Gamma = (r - 1)[C_{out} - C_{buy} + (N - 1)\beta]/N^2 > 0$ is the marginal value of uncertainty for unanimity. The belief threshold above which majority dominates is $\bar{\mu} = 1 - |D(1)|/\Gamma$. Numerically, $\bar{\mu}$ is nearly stable: for $N = 164$ and $\beta = 0.9$, increasing α from 0.05 to 0.49 (an 18-fold increase beyond α^*) only lowers $\bar{\mu}$ from 0.87 to 0.71 when $r = 1.5$. The mechanism fails only near certainty about the gains from cooperation, precisely where private information has least value (Figure 4 in Section 9.2 maps the full (α, μ) frontier).

8.2 Formation sets and institutional comparison

Let $\mathcal{F}_R \equiv \{p \in (0, 1] : V_W^{R1}(p, R) \geq c\}$ denote the *formation set* under rule R : the set of priors at which the institution forms. Since weak states' aggregate payoff is weakly lower under unanimity whenever the hegemon's payoff is higher (Theorem~1), the unanimity entry constraint is harder to satisfy; hence $\mathcal{F}_U \subseteq \mathcal{F}_M$ (Appendix B.6). The institutional comparison is therefore determined by the entry margin: unanimity dominates when the institution forms, but majority can dominate when only it can induce entry.

Corollary 1 (Unanimity dominance on the formation set). *Suppose $\alpha < \alpha^*(N, \beta)$. For every prior*

$$p \in \mathcal{F}_U,$$

$$V_H(U, p) > V_H(M, p).$$

Moreover, $\mathcal{F}_U \subseteq \mathcal{F}_M$.

Proof. See Appendix B.6. □

The argument combines budget balance with Theorem 1. Under majority, agreement occurs without delay and the pie is divided exactly among N players, so the hegemon's and weak states' expected payoffs sum to $V_e(\mu)$. Under unanimity, rejection on the aggressive branch destroys surplus through discounting, so the same budget constraint holds as a weak inequality. Since the hegemon captures strictly more under unanimity (Theorem 1), weak states must receive strictly less at every belief. The entry threshold is therefore higher under unanimity, and any prior that sustains entry under unanimity also sustains it under majority.

Remark 2 (Weak states prefer majority). *Under the conditions of Theorem 1, $V_W(\mu, M) > V_W(\mu, U)$ for every $\mu \in (0, 1]$, and $\mathcal{F}_U \subseteq \mathcal{F}_M$. Unanimity redistributes from weak states to the hegemon; it is never Pareto-improving conditional on entry.*

Why, then, would weak states participate under unanimity? Because the hegemon chooses the voting rule. A weak state that refuses to enter under unanimity does not get majority instead—it gets no institution at all, forfeiting whatever surplus the organization would generate. Since weak states are ex ante identical and decide simultaneously, the symmetric equilibrium is all-or-nothing: either all enter or none does (Section 4). There is no scope for individual holdout to force a rule change. Conditional on the hegemon's choice of unanimity, entry is individually rational whenever $V_W(p, U) \geq c$. The irony is that the veto, which guarantees each weak state inclusion in every agreement, is precisely what creates the screening problem that benefits the hegemon: by guaranteeing inclusion, unanimity forces weak proposers to make offers without knowing the hegemon's type, generating the overpayment that drives the conditional payoff advantage.

The following result completes the institutional comparison by classifying all priors.

Proposition 4 (Institutional classification). *Suppose $\alpha < \alpha^*(N, \beta)$. The hegemon's preferred rule*

is characterized as follows:

$$\begin{aligned}
U \succ M & \text{ if and only if } p \in \mathcal{F}_U, \\
M \succ U & \text{ if and only if } p \in \mathcal{F}_M \setminus \mathcal{F}_U, \\
U \sim M & \text{ if and only if } p \notin \mathcal{F}_M.
\end{aligned}$$

Since $\mathcal{F}_U \subseteq \mathcal{F}_M$, majority's only advantage is that it can support institutional entry at priors for which unanimity cannot.

Proof. See Appendix B.8. □

When the prior is such that weak states find it individually rational to enter under unanimity, the hegemon strictly prefers it (Corollary~1). When only majority can sustain entry, majority dominates through wider institutional viability rather than better bargaining terms. When neither rule can induce entry, the voting rule is irrelevant. When entry costs are low enough that $\mathcal{F}_U = (0, 1]$, unanimity dominates at every prior.

Remark 3 (Supermajority, weighted voting, and strategic inclusion). *The qualitative result is robust to the choice of majority threshold. Under any unweighted supermajority quota $q < N$, the exclusion logic is unchanged: weak proposers can still assemble a winning coalition without the hegemon, and no screening occurs. A higher quota reduces λ_M (because the hegemon must buy more coalition partners when it proposes), which has two consequences. First, the hegemon's payoff under majority falls, so the dominance threshold α^* increases with q : the condition $\alpha < \alpha^*$ becomes easier to satisfy. Second, by budget balance, weak states' payoff under majority rises, expanding the majority entry set $\mathcal{F}_M$. The unanimity entry set $\mathcal{F}_U$ is unaffected, since unanimity payoffs do not depend on the majority quota. In the (p, c) plane of Figure 3, this means the orange region (majority dominates via entry) expands at the expense of the white region (no institution forms), while the blue region ($\mathcal{F}_U$) remains fixed. Supermajority makes majority more accessible to weak states, but also makes it less attractive to the hegemon relative to unanimity, without altering the qualitative structure of the comparison.*

The more interesting extension is weighted voting. Consider weighted majority with weight $w_H > 1$

for the hegemon and weight 1 for each weak state, where the total weak-state weight still suffices to pass proposals without the hegemon. When w_H is small, including the hegemon in a winning coalition costs more than buying w_H additional weak states: $V_H^{R2}(\theta) > w_H \cdot V_W^{R2}(\mu)$ for all (θ, μ) . In this regime, weak proposers exclude the hegemon exactly as under simple majority, and the payoff remains $\lambda_M(w_H) \cdot V_e(\mu)$, with λ_M increasing in w_H because the hegemon needs fewer coalition partners when it proposes. No screening occurs. However, if w_H exceeds a threshold w_H^* at which buying the hegemon's vote becomes cheaper than assembling an equivalent coalition of weak states, weak proposers rationally include the hegemon even though its vote is not formally required. At that point, the proposer faces the same screening problem as under unanimity: choosing between an aggressive offer accepted by all types and a conservative offer rejected by $\theta = 1$. Screening thus depends on strategic inclusion, not on formal pivotality. One may thus interpret the unanimity–majority comparison in this paper as two polar cases of this continuum.

Remark 4 (Information design). The model also clarifies why information design would matter differently under the two voting rules. Under majority rule, the hegemon's continuation payoff is affine in weak states' posterior belief once entry is feasible. Splitting beliefs therefore does not create additional value beyond the payoff implied by the prior. Under unanimity, by contrast, weak states must include the hegemon in any agreement. This creates a screening discontinuity at the posterior threshold at which entry becomes feasible. As in standard Bayesian persuasion [Kamenica2011Bayesian], such nonconcavities are precisely where information design can matter. Thus, although persuasion is not necessary for the paper's main result, the model identifies why consensus institutions are more hospitable to informational leverage than majority institutions.

Example 2 (The full institutional comparison). Continuing with the parameters from Example 1 ($N = 13$, $r = 1.5$, $\alpha = 0.19$, $\beta = 0.9$), suppose entry costs $c = 0.07$. Under majority, the institution forms for $p \gtrsim 0.17$: $\mathcal{F}_M \approx [0.17, 1]$. Under unanimity, the screening mechanism reduces weak states' surplus, so the institution forms only for $p \gtrsim 0.38$: $\mathcal{F}_U \approx [0.38, 1]$.

For $p \in \mathcal{F}_U$ (above ≈ 0.38), unanimity strictly dominates: at $p = 0.50$, the hegemon's payoff under unanimity exceeds majority by 23%. For $p \in \mathcal{F}_M \setminus \mathcal{F}_U$ (between ≈ 0.17 and ≈ 0.38), majority dominates because only it can sustain entry. The advantage of majority at pessimistic priors is entirely through the entry channel: conditional on participation, unanimity always gives

the hegemon more (Theorem 1).

8.3 Numerical characterization

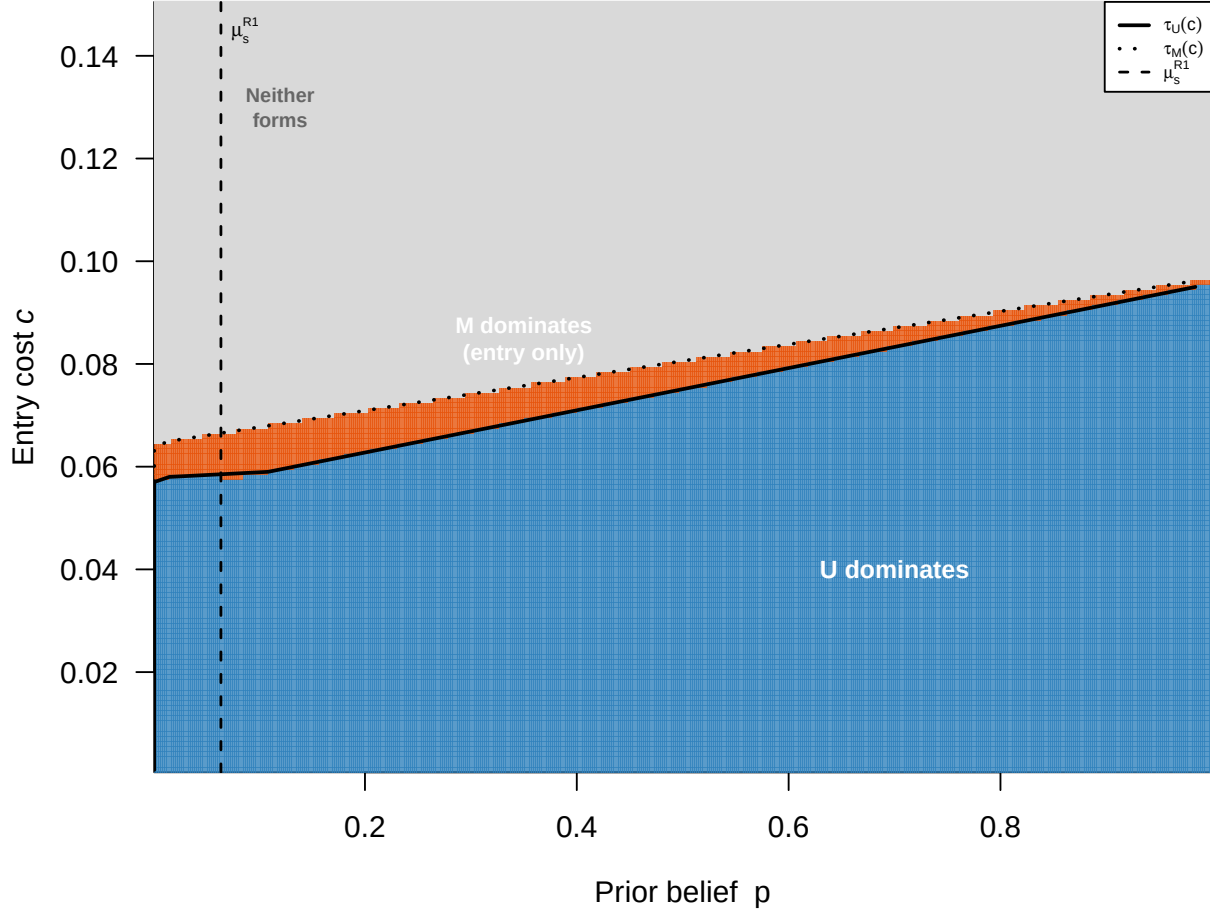


Figure 3: Institutional classification in the (p, c) plane ($N = 13$, $r = 1.5$, $\alpha = 0.19$, $\beta = 0.9$). Blue indicates pairs for which unanimity sustains entry and therefore dominates majority conditional on participation. Orange indicates pairs for which majority sustains entry but unanimity does not, so majority dominates through the entry channel. Gray indicates pairs for which neither rule sustains entry. The vertical dashed line marks the screening cutoff μ_s^{R1} . At this cutoff, weak proposers switch from aggressive to conservative offers under unanimity, generating an upward jump in the hegemon's payoff and, by budget balance, a downward jump in weak-state payoffs. This can make the unanimity formation set disconnected.

Figure 3 maps the institutional classification directly. For each combination of prior p and entry cost c , the figure shows which rule the hegemon prefers: unanimity (blue), majority through the entry channel (orange), or neither (gray). The unanimity-dominance region is always nested inside the

majority-entry region, confirming that $\mathcal{F}_U \subseteq \mathcal{F}_M$. The unanimity-dominance region expands with r (larger screening rents from greater type dispersion) and β (more patient bargaining strengthens the off-path threat that sustains screening), and contracts with α (stronger outside options raise the cost of buying unanimous agreement) and c (higher entry costs shrink $\mathcal{F}_U$ faster than $\mathcal{F}_M$, widening majority's scope advantage). The figure also illustrates why entry under unanimity need not be monotone in the prior. The same screening rent that gives the hegemon a conditional advantage under unanimity lowers weak-state payoffs at the cutoff, creating a local failure of entry under unanimity even when entry is feasible at lower and higher priors.

9 Discussion

9.1 Illustration: OPEC and the 1985–86 Price War

OPEC illustrates the screening mechanism in a setting where the institutional mapping is unusually clean. The hegemon is Saudi Arabia, the dominant producer with approximately 60% of global spare capacity. The weak states are the remaining OPEC members—Iran, Iraq, Nigeria, Venezuela, and others—who collectively depend on Saudi restraint to sustain cartel rents. The cooperation surplus $V(\theta)$ is the monopoly rent from collectively restricting supply above the competitive level. The state of the world θ is Saudi Arabia's true spare production capacity: how much additional oil Saudi Arabia can bring to market within 90 days. This capacity determines the value of the cartel agreement for all members, because it governs the credibility and costliness of the punishment that follows a breakdown. Saudi Arabia's outside option α is unilateral production—selling at market prices through bilateral contracts with China, India, and other importers—which yields substantial revenue even without OPEC coordination, given Saudi extraction costs of approximately \$3 per barrel, the lowest in the world (Alhajji and Huettner 2000; Nakov and Nuño 2013). The decision-making rule is unanimity: Article 11 of the OPEC Statute requires that “all decisions of the Conference, other than on procedural matters, shall require the unanimous agreement of all Full Members.”

The informational asymmetry at the core of the model is well documented. Saudi Arabia's true spare capacity has been closely guarded since 1982, when Oil Minister Yamani discontinued field-

by-field production reporting to industry periodicals (Simmons 2005). No independent audit of Saudi reserves was conducted until 2019, when DeGolyer & MacNaughton was engaged ahead of the Aramco IPO. Expert disagreement about actual deployable capacity is substantial: the IEA estimates Saudi spare capacity at approximately 2.4 million barrels per day, while independent analysts place the figure between 600,000 and 1 million, a divergence of 40–80% (Fattouh and Mahadeva 2013). In December 2025, the U.S. Energy Information Administration revised its methodology to distinguish “theoretical” from “effective” production capacity, implicitly acknowledging that much of OPEC’s reported spare capacity may exist only on paper. Other OPEC members cannot independently verify Saudi claims, and this uncertainty is precisely what gives Saudi Arabia informational leverage: when weaker members must decide how much to concede in quota negotiations, they do not know how costly a Saudi-led price war would actually be.

The December 1986 OPEC agreement illustrates the screening mechanism. By mid-1985, endemic quota cheating by other members had driven Saudi Arabia’s production share to roughly 14% of OPEC output, and its revenue had fallen below the Cournot equilibrium payoff—the level it could guarantee unilaterally by producing at capacity without any cartel agreement (Griffin and Neilson 1994). The price collapse of 1985–86, in which spot prices fell from \$28 to \$8.55 per barrel, restored the conditions for cooperation by making the cost of non-cooperation vivid for all members (Griffin and Neilson 1994; Yergin 1991).

With cooperation reestablished under OPEC’s unanimity rule, Saudi Arabia proposed the terms of a new agreement in December 1986: a price target of \$18 per barrel, a production ceiling of 15.8 million barrels per day, and a guaranteed quota share of approximately 25% for Saudi Arabia, nearly double its mid-1985 share. The other members accepted and, for the first time, began to comply with quotas in practice. This maps onto the branch of the model where the hegemon proposes in Round 1. The rent extraction does not come from the proposal itself. In a single-round ultimatum, any proposer captures the entire surplus. It stems from the continuation game: rejecting the hegemon’s offer leads to further bargaining rounds in which weak states face the screening problem created by uncertainty about Saudi Arabia’s true spare capacity. This continuation threat is what depresses weak states’ reservation values under unanimity relative to those under majority. Members did not know how long Saudi Arabia could sustain low prices or how costly renewed confrontation would

be. This uncertainty depressed their effective reservation value, allowing Saudi Arabia to secure a larger share than a transparent bargaining environment would have produced. Under a hypothetical majority rule, other members could have set quotas without Saudi consent, bypassing the screening problem entirely—but they would also have lost the enforcement mechanism that only Saudi spare capacity provides.

Recent events reinforce this interpretation. Angola left OPEC in January 2024 after its quota was cut by 350,000 barrels per day; its oil minister stated that OPEC was “not serving the country’s interests.” The UAE exited in April 2026, citing a persistent gap between its production capacity of approximately 5 million barrels per day and its assigned quota of 3.2 million. Both departures are consistent with the model’s institutional classification (Proposition~4): as a member’s outside option improves—Angola developing non-oil sectors, the UAE investing in capacity expansion—the formation set $\mathcal{F}_U$ contracts, and unanimity ceases to be individually rational. OPEC’s response in 2027, commissioning independent capacity audits by DeGolyer & MacNaughton for 19 member states, is a direct attempt to reduce the informational asymmetry that sustains screening rents. If successful, it would narrow the gap between aggressive and conservative offers, reducing Saudi Arabia’s informational advantage but potentially stabilizing the coalition by making quota allocation more transparent.

The OPEC case has limitations as an illustration of the static model. The repeated-game structure matters: Saudi Arabia’s credibility in 1986 depended on its demonstrated willingness to sustain a price war, and this reputation has shaped subsequent negotiations (Griffin and Neilson 1994). The model captures the structure of each bargaining episode but not the reputational dynamics that link them. Weak states within OPEC are heterogeneous, whereas the model assumes symmetric weak states. Unanimity is already in place at the time of the events analyzed, so we effectively are conditioning on entry.

9.2 Scope

The model isolates the effect of voting rules by holding proposal rights symmetric. This is not meant to deny that powerful states often exercise agenda control. Rather, agenda control creates a different problem: if the hegemon were the exclusive proposer and weak states had no bargaining

outside option ($d_W = 0$), weak states' expected bargaining payoff would be zero. With any positive entry cost $c > 0$, they would not join, and the institution would fail to form. Monopoly proposal power is therefore self-defeating when participation is voluntary. The relevant institutional choice is between voting rules that can sustain participation.

The comparison is therefore between unanimity and symmetric simple majority. Majority rule here does not grant the hegemon agenda control, weighted voting, or privileged coalition-building capacity. This choice isolates the effect of the voting rule on information use, holding other institutional features constant. In practice, majority rule in international organizations is often accompanied by weighted voting or asymmetric proposal rights, as in the IMF or World Bank. Those features may benefit powerful states independently. Remark 3 considers this possibility by arguing that screening still works whenever coalitions strategically include the hegemon, whether or not formal rules make its vote pivotal.

The hegemon's outside option is proportional to the value of cooperation: $d_H = \alpha V(\theta)$. This assumption captures the idea that the same conditions that make multilateral cooperation valuable also improve the hegemon's bilateral fallback. The proportional form separates informational power, which depends on θ being unknown to weak states, from bargaining power, which depends on α . The restriction $\alpha < 1/r$ ensures that multilateral cooperation remains strictly more valuable than the bilateral alternative in every state.

The all-or-nothing entry assumption serves the same isolating purpose. The parameter N is the size of the institution the hegemon proposes to create, not the total number of states in the international system. The hegemon invites a specific group of $N - 1$ weak states, and the institution forms only if all join. This suppresses sequential accession dynamics such as free-riding, learning from early entrants, and collective action problems among weak states. Those dynamics could matter, but they would introduce mechanisms that operate in opposing directions. The model focuses on the screening channel; characterizing its interaction with sequential entry remains open.

The binary type space is also less restrictive than it first appears. Binary types provide the cleanest way to show how pivotality creates screening. Appendix C shows that the screening rent under unanimity is strictly positive for any continuous distribution with full support, while majority eliminates it entirely. For the uniform distribution, the dominance threshold satisfies $\alpha_{\text{cont}}^* \geq \alpha^*$, suggesting

that binary types are the most conservative specification for the mechanism rather than an optimistic modeling choice.

The mechanism is most relevant when four conditions hold: informational asymmetry is substantively important (H knows θ , while weak states do not); the hegemon has a valuable outside option ($\alpha > 0$); entry is costly and belief-dependent; and the voting rule makes the hegemon pivotal, either formally or through strategic inclusion.

Figure 4 maps the conditional dominance region in the (α, μ) plane. Blue regions mark beliefs at which unanimity gives the hegemon a higher continuation payoff; red regions mark beliefs at which majority dominates. The solid curve traces the frontier $\bar{\mu}(\alpha)$ from Remark 1. Two patterns stand out. First, the frontier is nearly flat: increasing α well beyond α^* shifts $\bar{\mu}$ only modestly downward, confirming that the screening advantage persists across most of the belief space. Second, the red region clusters near $\mu = 1$, where weak states are nearly certain about the gains from cooperation. This is precisely where informational asymmetry has the least value, and the cost of buying unanimous agreement dominates.

Finally, the scope condition $\alpha < \alpha^*$ should be read as the region in which majority can never dominate through a higher conditional-on-entry payoff. In that region, any advantage of majority must operate through entry: majority may sustain participation at priors for which unanimity does not, because $\tau(U) > \tau(M)$. This is most likely when entry costs are high, when α approaches α^* , or when information is nearly public.

The boundary is sharp. At $\alpha = \alpha^*$, the payoff difference $D(\mu)$ equals zero exactly at $\mu = 1$. Because α^* decreases with N , the condition is more demanding in large organizations, where unanimity requires buying many more votes than majority. For $\alpha > \alpha^*$, majority can dominate near $\mu = 1$, where weak states nearly know the hegemon's type and the cost of buying unanimous agreement outweighs the screening advantage. Away from that near-certainty region, unanimity can still dominate. Figure 4 illustrates this pattern: even in the $N = 164$ panels, the blue region covers most of the (α, μ) plane, while the red region clusters near $\mu = 1$.

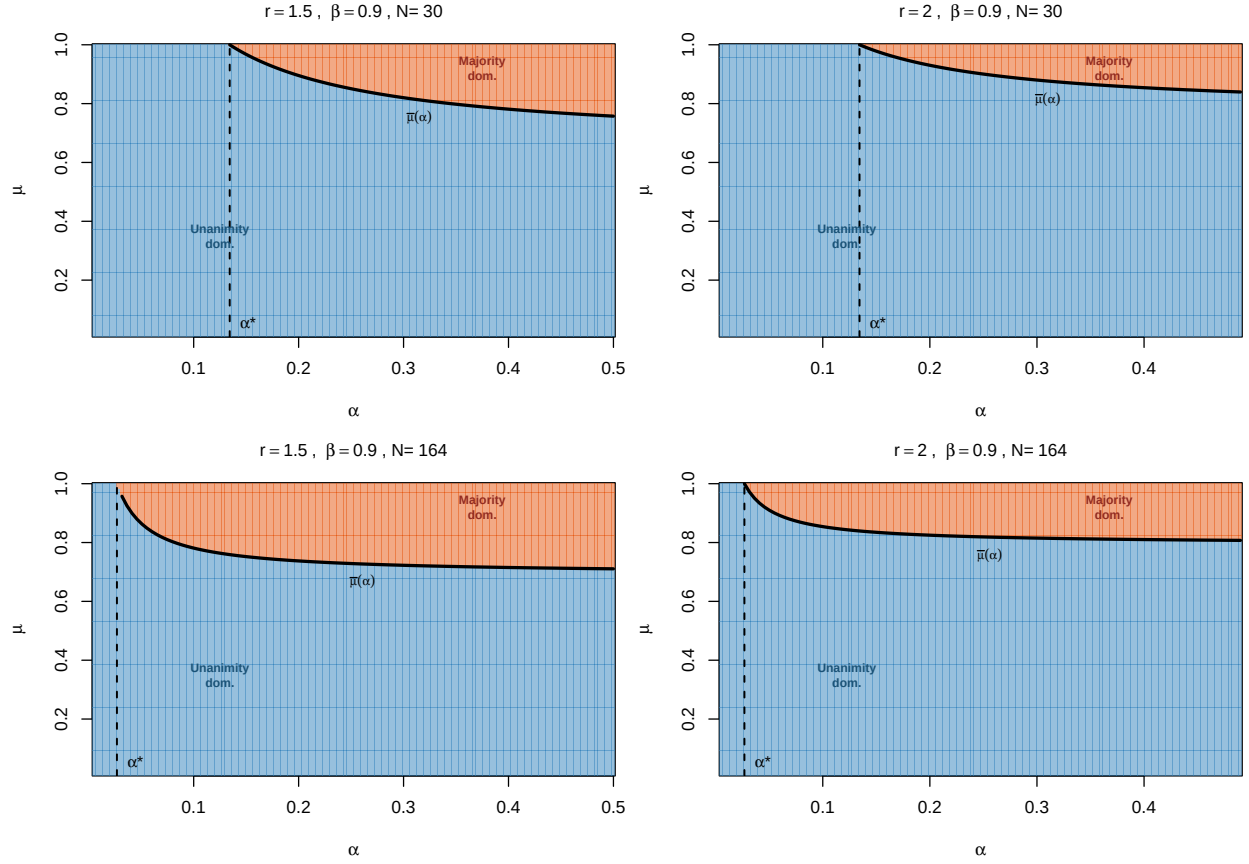


Figure 4: Conditional payoff dominance in the (α, μ) plane. Blue: unanimity dominates ($D(\mu) > 0$). Red: majority dominates ($D(\mu) < 0$). Solid curve: belief threshold $\bar{\mu}(\alpha)$ above which majority dominates (Remark~1). Dashed line: α^* . Left column: $r = 1.5$. Right column: $r = 2$. Top row: $N = 30$. Bottom row: $N = 164$. All panels: $\beta = 0.9$.

10 Conclusion

This paper has identified conditions under which consensus is a rational choice for a hegemon. The key is to distinguish between two modes of exercising power under different voting rules. Majority rule with symmetric proposals allows weak states to bypass the hegemon, thereby eliminating the screening problem. Unanimity forces weak states to include the hegemon under uncertainty, creating a screening cutoff that gives the hegemon a conditional payoff advantage under the conditions characterized above. The institutional comparison has a clean structure: unanimity dominates wherever it can sustain entry, and majority's only advantage is wider institutional viability.

The argument does not imply that all consensus institutions are designed for this reason, or that informal power disappears under consensus. It shows something more specific: under identifiable conditions on informational asymmetry and the prospects of cooperation, unanimity becomes the preferred institutional arrangement for a hegemon. Consensus may be less a concession by power than a way of exercising power through a different channel: the channel of information.

The policy implication is that formal equality under consensus can coexist with asymmetric bargaining power. In organizations where the value of cooperation is difficult to assess, consensus may protect participation while also increasing the rents of actors with superior information. Reforms that improve transparency, independent technical assessment, and the analytical capacity of weaker members should weaken this informational advantage more directly than changing the voting rule alone.

The main results use a binary type space. It is important to distinguish what generalizes from what does not. The screening rent under unanimity is strictly positive for any continuous distribution on $[1, r]$ with full support (Appendix C). Under majority, W bypasses H , and the rent is zero. This structural contrast—screening under unanimity, no screening under majority—is distribution-free. What does not generalize without qualification is the dominance threshold α^* . For binary types, Theorem 1 provides a sharp biconditional. For the uniform distribution on continuous types, the dominance threshold α_{cont}^* is weakly larger than α^* , suggesting that binary types are the most conservative case. For other distributions, the threshold has not been characterized. A general characterization of α^* for arbitrary distributions remains an open question.

Several extensions remain. First, if weaker states can invest in independent analytical capacity, informational power erodes endogenously over time. Second, heterogeneous outside options across weak states would enrich the screening dynamics by introducing middle powers. Third, repeated play within an established consensus institution could generate learning that narrows the informational gap. These directions point toward a richer theory of how informational power rises and declines within institutional frameworks.

Appendix A: Bargaining Derivations

Notation summary

Symbol	Meaning	Defined in
N	Number of players (1 hegemon + $N - 1$ weak states)	Def. 1
$\theta \in \{0, 1\}$	State of the world	Def. 1
p	Prior $\Pr(\theta = 1)$	Def. 1
μ	Posterior belief $\Pr(\theta = 1 \mid \text{history})$; equals p at entry, updated after R1 rejection	Sec. 3
$V(\theta)$	Value of cooperation: $V(0) = 1$, $V(1) = r$	Def. 1
$r > 1$	High-type value	Def. 1
$\alpha \in (0, 1/r)$	Outside-option share: $d_H = \alpha V(\theta)$	Def. 1
$\beta \in (0, 1)$	Common discount factor	Def. 1
$c > 0$	Entry cost per weak state	Def. 1
$V_e(\mu)$	Expected cooperation value $1 + \mu(r - 1)$	Sec. 3
x	Shorthand $(N - 1)\alpha r$	Sec. 3
q	Majority quota $\lfloor N/2 \rfloor + 1$	Def. 1
$R \in \{M, U\}$	Voting rule (majority / unanimity)	Def. 1
μ_s^{R2}	R2 screening cutoff	App. A.2
μ_s^{R1}	Superseded single-cutoff object; replaced by constrained A/C/R regimes	App. A.3–A.4
ϕ	Superseded auxiliary for the old single-cutoff formula	App. A.3–A.4
$\tau(R)$	Entry threshold under rule R	Sec. 6
$v(\mu, R)$	Hegemon's net gain from institution: $E[V_H^{R1}] - \alpha V_e(\mu)$ if entry, 0 otherwise	Sec. 6
$\bar{\alpha}$	Regime threshold for α -independent R1 cutoff	Prop. 2
α^*	Threshold for conditional dominance (Thm. 1): necessary and sufficient	Eq. (4)
$\omega(\mu)$	R2 continuation shorthand: $(N - 2)\beta V_W^{R2}(\mu)$	App. A.3
κ_M^E	Corrected majority weak-state coefficient: $[N(N - 1) + \beta(q - 1)]/[N^2(N - 1)]$	App. A.1
$D(\mu)$	Conditional payoff difference $V_H^{R1}(\mu, U) - V_H^{R1}(\mu, M)$	Thm. 1
Γ	Marginal value of uncertainty for unanimity	Remark 1
$\bar{\mu}$	Belief threshold above which majority dominates: $1 - D(1) /\Gamma$	Remark 1
$C_{\text{buy}}, C_{\text{out}}$	Vote-buying cost, outside-option cost (proof)	App. B.5
λ_M^E	Corrected majority payoff coefficient	App. A.1–B.1
$\mathcal{F}_R$	Formation set under rule R : $\{p : V_W^{R1}(p, R) \geq c\}$	Sec. 7

A.1 Majority: R2 and R1 under external outside options

.0.0.1 Status. Verified under the corrected accounting convention. When a weak proposer excludes H under majority, H receives $\alpha V(\theta)$ from an outside option external to the institutional pie. This payoff is not paid by the weak coalition.

Under majority, weak proposers can form a winning coalition from other weak states. Since including H requires a strictly positive transfer and excluding H does not reduce the institutional pie available to the weak coalition, weak proposers exclude H in both rounds. The hegemon's private information therefore does not create a screening problem under majority.

R2 continuation values:

$$V_H^{R2}(\theta, M) = \frac{V(\theta)[1 + (N - 1)\alpha]}{N}, \quad (5)$$

$$V_W^{R2}(\mu, M) = \frac{V_e(\mu)}{N}. \quad (6)$$

The previous expression $V_W^{R2}(\mu, M) = (1 - \alpha)V_e(\mu)/N$ treated H 's outside option as if it were subtracted from the weak coalition's pie. That expression is not consistent with the model's external outside-option interpretation.

In R1, if H proposes, it buys $q - 1$ weak votes at cost $\beta V_W^{R2}(\mu, M)$ each. If a weak state proposes, it excludes H and H receives $\alpha V(\theta)$ externally. Thus:

$$\begin{aligned} E_\theta[V_H^{R1}(\theta, \mu, M)] &= \frac{1}{N} \left[V_e(\mu) - (q - 1)\beta \frac{V_e(\mu)}{N} \right] + \frac{N - 1}{N} \alpha V_e(\mu) \\ &= \lambda_M^E V_e(\mu), \end{aligned} \quad (7)$$

where

$$\lambda_M^E \equiv \frac{N[1 + (N - 1)\alpha] - \beta(q - 1)}{N^2}.$$

The representative weak-state payoff is:

$$V_W^{R1}(\mu, M) = \kappa_M^E V_e(\mu), \quad \kappa_M^E = \frac{N(N - 1) + \beta(q - 1)}{N^2(N - 1)}. \quad (8)$$

Finally,

$$\lambda_M^E > \alpha \iff \alpha < 1 - \frac{\beta(q-1)}{N}.$$

This condition is not automatic and must be imposed whenever the comparison requires majority bargaining to dominate the outside option.

A.2 Unanimity: R2

.0.0.2 Status. Verified, up to tie-breaking at the cutoff.

When H proposes in R2, it offers zero to all weak states and keeps $V(\theta)$. When W proposes, the proposer compares an aggressive offer, accepted only by the low type, with a conservative offer, accepted by both types:

$$g(\mu) \equiv \max\{(1-\mu)(1-\alpha), V_e(\mu) - \alpha r\}.$$

The R2 weak-state continuation value is:

$$W_2(\mu) \equiv V_W^{R2}(\mu, U) = \frac{g(\mu)}{N}. \quad (9)$$

The screening cutoff is:

$$\mu_s^{R2} = \frac{\alpha(r-1)}{r-\alpha}. \quad (10)$$

Below the cutoff, W plays aggressive; above it, W plays conservative. At μ_s^{R2} , W is indifferent and the equilibrium requires an explicit tie-breaking convention.

The hegemon's R2 continuation values under unanimity are:

$$V_H^{R2}(\theta = 1, \mu) = \frac{r[1 + (N-1)\alpha]}{N} \quad (\text{constant in } \mu), \quad (11)$$

$$V_H^{R2}(\theta = 0, \mu < \mu_s^{R2}) = \frac{1 + (N-1)\alpha}{N}, \quad (12)$$

$$V_H^{R2}(\theta = 0, \mu > \mu_s^{R2}) = \frac{1 + (N-1)\alpha r}{N} \quad (\text{overpaid}). \quad (13)$$

Equivalently, weak states' R2 continuation values are:

$$V_W^{R2}(\mu < \mu_s^{R2}) = \frac{(1 - \mu)(1 - \alpha)}{N}, \quad (14)$$

$$V_W^{R2}(\mu > \mu_s^{R2}) = \frac{V_e(\mu) - \alpha r}{N}. \quad (15)$$

Continuity at μ_s^{R2} is verified by direct substitution.

A.3 Unanimity: R1 when a weak state proposes

.0.0.3 Status. Verified under strict BF feasibility. The previous single-cutoff derivation is superseded.

Let

$$A_0 \equiv 1 + (N - 1)\alpha, \quad A_1 \equiv 1 + (N - 1)\alpha r, \quad k \equiv N - 2.$$

When a weak state proposes in R1 under unanimity, it must secure H 's vote and the votes of the k non-proposing weak states. Since the weak proposer does not know θ , any accepted proposal that can pass in the low state must fit inside the low-state pie.

.0.0.4 Conservative offer. A conservative offer is accepted by both types of H . The high type binds, so the transfer to H is:

$$h_C = \frac{\beta r A_0}{N}.$$

Each non-proposing weak state receives its discounted R2 continuation value, $\beta W_2(\mu)$. The conservative proposer payoff is:

$$C(\mu) = V_e(\mu) - h_C - k\beta W_2(\mu), \quad (16)$$

and the offer is feasible iff:

$$h_C + k\beta W_2(\mu) \leq 1. \quad (17)$$

.0.0.5 Aggressive offer. An aggressive offer is accepted by the low type and rejected by the high type. Because a high-type rejection is public and leads to posterior one in R2, the low type must

receive:

$$h_A = \frac{\beta A_1}{N}.$$

Under standard BF voting with public vote profiles, a non-proposing weak state is pivotal only when H accepts, i.e., in the low state. Thus its binding transfer is:

$$y_A = \beta W_2(0) = \frac{\beta(1 - \alpha)}{N}.$$

The aggressive proposer payoff is:

$$A(\mu) = (1 - \mu) [1 - h_A - ky_A] + \mu\beta W_2(1), \quad (18)$$

and the offer is feasible iff:

$$h_A + ky_A \leq 1. \quad (19)$$

.0.0.6 Deliberate rejection. The proposer can also make an offer that both types reject, yielding:

$$R(\mu) = \beta W_2(\mu). \quad (20)$$

Therefore, the verified R1 weak-proposer value is:

$$W_1^{prop}(\mu, U) = \max\{A(\mu) \text{ if feasible, } C(\mu) \text{ if feasible, } R(\mu)\}. \quad (21)$$

This is the key correction. Strict BF feasibility can remove the conservative option at high beliefs, so R1 screening is not generally represented by a single monotone aggressive-to-conservative cutoff.

A.4 Unanimity: R1 when the hegemon proposes

.0.0.7 Status. Partly verified. The old pooling formula is valid only on the pooling branch. A selection-free lower bound is verified and used below; a complete pure-strategy PBE characterization outside the pooling region remains pending.

The old H-proposer expression assumes that both types of H offer each weak state $\beta W_2(\mu)$ and

that the offer is accepted. This pooling expression,

$$V(\theta) - (N - 1)\beta W_2(\mu),$$

is valid only if the low-state proposal is feasible and both types prefer agreement to rejection.

A verified selection-free lower bound is:

$$L_H(\mu) = (1 - \mu)\frac{\beta A_0}{N} + \mu \left[r - (N - 1)\frac{\beta r(1 - \alpha)}{N} \right]. \quad (22)$$

To see this, the low type can offer zero. If the offer is accepted, it receives $1 \geq \beta A_0/N$; if it is rejected, its R2 continuation is at least $\beta A_0/N$. The high type can offer each weak state $\beta W_2(1) + \varepsilon$. Since $W_2(q) \leq W_2(1)$ for all posteriors q , this offer is accepted for any beliefs. Letting $\varepsilon \downarrow 0$ gives the high-type term in (22).

A.5 Calibrated R1 weak-proposer regimes

.0.0.8 Status. Verified for the calibration $N = 13$, $r = 1.5$, $\alpha = 0.19$, $\beta = 0.9$.

For this calibration:

$$\mu_s^{R2} = 0.072519, \quad h_C = 0.340615, \quad h_A = 0.306000, \quad y_A = 0.056077.$$

The aggressive offer is feasible:

$$h_A + 11y_A = 0.922846 < 1.$$

The aggressive and conservative payoffs cross at:

$$\mu_{AC} = 0.031188.$$

The conservative offer is feasible only up to:

$$\mu_C^F = 0.301717.$$

Hence the weak-proposer regime is:

$$A \quad \text{for} \quad \mu < 0.031188,$$

$$C \quad \text{for} \quad 0.031188 < \mu \leq 0.301717,$$

$$A \quad \text{for} \quad \mu > 0.301717.$$

Here A means the aggressive offer and C means the conservative offer. The final return to A is not a new screening cutoff; it is caused by strict low-state feasibility. At high beliefs, the conservative offer would require transfers that do not fit inside the low-state pie.

A.6 Budget checks

.0.0.9 Status. The old budget-balance argument is superseded.

Under corrected majority accounting,

$$E[V_H(\mu, M)] + (N - 1)V_W(\mu, M)$$

includes H 's external outside-option payoff when H is excluded. It is therefore not an identity for division of the institutional pie. Any proof of $\mathcal{F}_U \subseteq \mathcal{F}_M$ must compare weak-state payoffs directly or impose separate sufficient conditions; it cannot follow from the old budget-balance step.

A.7 Entry thresholds

.0.0.10 Status. Majority entry is verified. Unanimity entry must be rederived from the corrected R1 subgame and the H-proposer branch.

With collective all-or-nothing entry by weak states, majority forms iff:

$$V_W^{R1}(\mu, M) = \kappa_M^E V_e(\mu) \geq c,$$

where κ_M^E is defined in (8). Thus:

$$\mathcal{F}_M^E = \{\mu \in (0, 1] : \kappa_M^E[1 + \mu(r - 1)] \geq c\}.$$

Equivalently, with the usual truncation,

$$\tau_M^E = \frac{c/\kappa_M^E - 1}{r - 1}.$$

The unanimity entry set remains pending because the weak-state R1 payoff must incorporate the corrected weak-proposer regimes and the unresolved H-proposer signaling branch. The old closed-form conservative threshold from the superseded draft is therefore not used in the corrected proof architecture.

Appendix B: Proofs

.0.0.11 Correction note. This appendix records the proof status under the corrected model: external outside options under majority and strict BF feasibility in every proposal. Results marked as pending are not used as established results in the corrected proof architecture.

B.1 Proof of Proposition 1 (Majority produces no screening)

.0.0.12 Status. Verified after correcting the outside-option accounting.

In R2, if H proposes, it buys $q - 1$ weak votes at zero cost and keeps $V(\theta)$. If a weak state proposes, it can form a winning coalition with other weak states and exclude H . Since H 's outside option is external, exclusion does not reduce the weak coalition's institutional pie. Including H would require a strictly positive transfer, so exclusion is optimal.

Therefore:

$$V_H^{R2}(\theta, M) = \frac{[1 + (N - 1)\alpha]V(\theta)}{N}, \quad V_W^{R2}(\mu, M) = \frac{V_e(\mu)}{N}.$$

In R1, if H proposes, it pays $(q - 1)\beta V_e(\mu)/N$ to weak coalition partners. If a weak state proposes,

it again excludes H , so H receives $\alpha V(\theta)$ externally. Hence:

$$E_\theta[V_H^{R1}(\theta, \mu, M)] = \lambda_M^E V_e(\mu), \quad \lambda_M^E = \frac{N[1 + (N-1)\alpha] - \beta(q-1)}{N^2}.$$

This is affine in μ . Since weak proposers exclude H at every belief, they never choose between aggressive and conservative offers to H ; hence there is no screening cutoff under majority. $\square$

The representative weak-state payoff is:

$$V_W^{R1}(\mu, M) = \kappa_M^E V_e(\mu), \quad \kappa_M^E = \frac{N(N-1) + \beta(q-1)}{N^2(N-1)}.$$

The auxiliary inequality needed for the entry-comparison case in which majority forms but unanimity does not is:

$$\lambda_M^E > \alpha \iff \alpha < 1 - \frac{\beta(q-1)}{N}.$$

Unlike the old proof, this inequality does not follow automatically from $\alpha < 1/r$.

B.2 Corrected R1 weak-proposer characterization under unanimity

.0.0.13 Status. Verified.

Appendix A.3 derives the three feasible weak-proposer options in R1 under unanimity:

$$C(\mu) = V_e(\mu) - h_C - (N-2)\beta W_2(\mu), \quad h_C = \frac{\beta r[1 + (N-1)\alpha]}{N},$$

with feasibility condition

$$h_C + (N-2)\beta W_2(\mu) \leq 1;$$

$$A(\mu) = (1-\mu)[1 - h_A - (N-2)y_A] + \mu\beta W_2(1),$$

where

$$h_A = \frac{\beta[1 + (N-1)\alpha r]}{N}, \quad y_A = \frac{\beta(1-\alpha)}{N},$$

with feasibility condition

$$h_A + (N-2)y_A \leq 1;$$

and the deliberate-rejection option

$$R(\mu) = \beta W_2(\mu).$$

Thus the weak proposer chooses:

$$W_1^{prop}(\mu, U) = \max\{A(\mu) \text{ if feasible, } C(\mu) \text{ if feasible, } R(\mu)\}.$$

This replaces the old proof of a unique monotone R1 cutoff. A single cutoff may exist on a subset where both accepted offers are feasible and dominate deliberate rejection, but it is not a global property of the strict BF subgame.

B.3 Selection-free H-proposer lower bound under unanimity

.0.0.14 Status. Verified. The complete H-proposer signaling subgame remains pending outside the pooling region, but the following lower bound does not rely on equilibrium selection.

For any posterior q ,

$$W_2(q) \leq W_2(1) = \frac{r(1 - \alpha)}{N}.$$

The low type of H can offer zero. If the offer is accepted, it receives 1; if rejected, it receives at least $\beta[1 + (N - 1)\alpha]/N$ in R2. Since $1 \geq \beta[1 + (N - 1)\alpha]/N$, the low type guarantees

$$\frac{\beta[1 + (N - 1)\alpha]}{N}.$$

The high type can offer each weak state $\beta W_2(1) + \varepsilon$, which is accepted under any posterior because $W_2(q) \leq W_2(1)$. Letting $\varepsilon \downarrow 0$, the high type guarantees

$$r - (N - 1) \frac{\beta r(1 - \alpha)}{N}.$$

Therefore the H-proposer branch is bounded below by:

$$L_H(\mu) = (1 - \mu) \frac{\beta[1 + (N - 1)\alpha]}{N} + \mu \left[r - (N - 1) \frac{\beta r(1 - \alpha)}{N} \right].$$

B.4 Calibrated lower-bound dominance check

.0.0.15 Status. Verified for the calibration $N = 13$, $r = 1.5$, $\alpha = 0.19$, $\beta = 0.9$.

For this calibration, $q = 7$ and

$$\lambda_M^E = 0.220355029586.$$

The verified weak-proposer regimes are:

$$A \text{ on } [0, 0.031188), \quad C \text{ on } (0.031188, 0.301717], \quad A \text{ on } (0.301717, 1].$$

When W proposes, H receives

$$H_A(\mu) = (1 - \mu)h_A + \mu h_C$$

on the aggressive branch and

$$H_C(\mu) = h_C$$

on the conservative branch, where

$$h_A = 0.306000, \quad h_C = 0.340615.$$

Define the lower-bound unanimity payoff:

$$U_{LB}(\mu) = \frac{1}{N}L_H(\mu) + \frac{N-1}{N}H_W(\mu),$$

where $H_W(\mu) = H_A(\mu)$ on the aggressive intervals and $H_W(\mu) = H_C(\mu)$ on the conservative interval. Majority is:

$$M(\mu) = \lambda_M^E V_e(\mu).$$

On each interval, $U_{LB}(\mu) - M(\mu)$ is affine, so endpoint checks suffice. The endpoint gaps are:

$$U_{LB}(0) - M(0) = 0.079574,$$

$$U_{LB}(0.031188) - M(0.031188) = 0.077767,$$

$$U_{LB}(0.301717) - M(0.301717) = 0.062089,$$

$$U_{LB}(1) - M(1) = 0.021621.$$

All are positive. Hence the lower bound on unanimity exceeds the corrected majority payoff for every $\mu \in [0, 1]$ in this calibration. This is a calibrated/parametric result, not a general theorem.

B.5 Pending: general conditional dominance theorem

.0.0.16 Status. Pending. The old proof of Theorem~1 used two objects that are no longer valid globally under strict BF:

1. the old majority coefficient with the factor $(1 - \alpha)$ in weak-state continuation values;
2. the old single-cutoff R1 unanimity formula.

The general theorem must be rederived either as a set of sufficient conditions based on the corrected piecewise regimes, or as a theorem based on the selection-free lower bound in Appendix B.3. Until that derivation is completed, Theorem~1 should not be treated as proven in its current global form.

B.6 Pending: formation-set nesting

.0.0.17 Status. Pending. The old proof of $\mathcal{F}_U \subseteq \mathcal{F}_M$ used a budget-balance identity under majority. With external outside options, majority payoffs include H 's outside-option payoff even when H is excluded, so that budget-balance step is invalid. The corrected proof must compare weak-state payoffs directly, using the corrected κ_M^E and the rederived unanimity weak-state payoff.

B.7 Pending: global maximum of weak-state payoff under unanimity

.0.0.18 Status. Pending. The old lemma used the old aggressive and conservative R1 payoff branches and the old H-proposer formula. It must be rechecked using the corrected weak-proposer regime $A/C/R$ and the H-proposer branch or a valid bound for weak-state payoffs.

B.8 Pending: institutional classification

.0.0.19 Status. Pending. The old classification requires three ingredients that are not yet all proven under the corrected model:

1. conditional dominance of unanimity on the relevant formation set;
2. $\mathcal{F}_U \subseteq \mathcal{F}_M$;
3. $\lambda_M^E > \alpha$ for the case in which majority forms and unanimity does not.

The third condition is equivalent to $\alpha < 1 - \beta(q - 1)/N$ and must be stated explicitly if that case is retained.

Appendix C: Continuous Types

.0.0.20 Status. Pending rederivation. The continuous-type appendix below was written under the superseded majority accounting convention and before the strict BF feasibility correction. It is retained in source as a draft, but it is hidden from the compiled appendix and should not be cited as a proven extension until majority payoffs, W-proposer feasibility, and H-proposer continuation values are rederived under the corrected model.

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